Genetic and molecular epidemiology of adult diffuse glioma

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Abstract | The WHO 2007 glioma classification system (based primarily on tumour histology) resulted in considerable interobserver variability and substantial variation in patient survival within grades. Furthermore, few risk factors for glioma were known. Discoveries over the past decade have deepened our understanding of the molecular alterations underlying glioma and have led to the identification of numerous genetic risk factors. The advances in molecular characterization of glioma have reframed our understanding of its biology and led to the development of a new classification system for glioma. The WHO 2016 classification system comprises five glioma subtypes, categorized by both tumour morphology and molecular genetic information, which led to reduced misclassification and improved consistency of outcomes within glioma subtypes. To date, 25 risk loci for glioma have been identified and several rare inherited mutations that might cause glioma in some families have been discovered. This Review focuses on the two dominant trends in glioma science: the characterization of diagnostic and prognostic tumour markers and the identification of genetic and other risk factors. An overview of the many challenges still facing glioma researchers is also included.

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Every year, ~100,000 people worldwide are diagnosed as having diffuse gliomas[1]. Although it comprises <1% of all newly diagnosed cancers, diffuse glioma is associated with substantial mortality and morbidity[2]. Glioblastoma, the most lethal glioma, accounts for 70–75% of all diffuse glioma diagnoses and has a median overall survival of 14–17 months. Globally, substantial differences in glioma incidence are evident between European and Asian populations[3], discussed further below. In addition to geographic differences, glioma incidence varies by age, sex, ethnicity and tumour histology whereas glioma survival varies by tumour subtype, age and sex.

Historically, gliomas were considered to originate from differentiated astrocytic and oligodendrocytic components of the CNS[4]. As such, the WHO 2007 classification of diffuse gliomas relied only on tumour histology[5]. Over the past decade, molecular studies of human tumours have provided important new insights into the complex genetic, chromosomal and epigenetic changes that accompany glioma formation and maintenance[6–11]. As a result, the WHO devised an integrated classification system, which was introduced in 2016 and was based on tumour morphology and molecular alterations[12]. Stem-like cells within the CNS are now thought to be the cells of origin of several primary brain tumour types, including glioblastoma[13]. Data obtained in animal models support the notion that neural stem cells[14], particularly in the subventricular zone, give rise to at least a subset of glioblastomas[15]. Many new genetic risk factors for gliomas were discovered during the past 10 years, including rare mutations confined to specific families and more common inherited variants in 25 independent genetic loci[16–19].

In this Review of adult diffuse gliomas, we focus on research advances that led from the WHO 2007 classification criteria to the WHO 2016 integrated classification system[12,20]. We present the latest incidence and survival data for adult diffuse glioma from several population-based sources, including the Central Brain Tumor Registry of the United States (CBTRUS) 2018 report[21], which relies on the WHO 2007 classification system as the WHO 2016 system has not yet been used widely enough. We also provide an update on clinical research, genetic and other risk factors for glioma and, where possible, relate these to the WHO 2016 glioma subgroups.

Classification of glioma

The 2007 WHO guidelines for classifying malignant gliomas were based on histological criteria, in which several morphological subtypes corresponded to the general appearance of the tissue of origin: astrocytoma, oligodendroglioma and mixed oligoastrocytoma. Tumour malignancy was graded II–IV according to morphological criteria and high-grade tumours generally had a poor

Key points

- Glioma incidence differs by age, sex, ethnicity and geography whereas glioma survival varies by tumour subtype, age and sex.
- In the past decade, multiple discoveries have expanded our understanding of glioma and led to a new classification system (WHO 2016) that integrates molecular alterations and histology.
- The WHO 2016 classification system defines five glioma subtypes that have improved homogeneity in their clinical outcomes.
- Twenty-five risk loci for glioma and several rare inherited mutations that might cause glioma in some families have been identified; however, ionizing radiation is the only confirmed environmental risk factor.

prognosis. However, this classification system had high interobserver variability[22–24] and survival varied substantially within grades. Therefore, neuropathologists and glioma biologists searched for markers to improve the characterization of clinically relevant subgroups[25].

In 2015, two research groups simultaneously published papers describing molecular and chromosomal subtypes that clarified glioma classification[26,27]. The first study included over 1,000 patients with WHO grade II–IV glioma from three large studies in the United States[27]. These researchers focused on the presence or absence of three tumour markers: promoter mutations in *TERT* (which encodes telomerase reverse transcriptase); mutations in *IDH1* or *IDH2* (encoding isocitrate dehydrogenase (NADP) cytoplasmic and mitochondrial subtypes, respectively[8,28–32], collectively referred to as *IDH* mutations), which had been shown to be common in some types of glioma and to be associated with survival[25]; and co-deletion of chromosome arms 1p and 19q, as observed in oligodendrogliomas[33]. These three markers classified patients with glioma into five groups, distinguished by differences in survival (especially between patients with grade II or III tumours), age at diagnosis, germline risk alleles, other tumour mutations and chromosome copy number changes. Notably, grade II or III astrocytomas with only *TERT* promoter mutations had outcomes very similar to glioblastoma. The second study included almost 300 patients with WHO grade II and III gliomas from The Cancer Genome Atlas (TCGA) and found three distinct subgroups based on *IDH* mutation, *TP53* mutation and 1p19q co-deletion status[26].

In 2016, building on data from these and other seminal studies[6], the WHO integrated tumour morphology, *IDH* mutation and 1p19q co-deletion status into a new classification system for adult diffuse glioma[12] (FIG. 1). This classification system includes five primary designations of adult diffuse glioma: glioblastoma, *IDH*-wild type; glioblastoma, *IDH*-mutant; diffuse or anaplastic astrocytomas, *IDH*-wild type; diffuse or anaplastic astrocytomas, *IDH*-mutant; and oligodendroglioma or anaplastic oligodendroglioma, *IDH*-mutant and 1p19q co-deleted (FIG. 1). Characterization of the third group was subsequently updated[20] as substantial evidence showed that many such tumours had WHO grade II or III histology but were associated with a clinical outcome similar to that of WHO grade IV tumours[8,27,34,35] (discussed below). A sixth group, based on histological appearance in the absence of molecular determination, is designated malignant glioma not otherwise specified. A seventh group, diffuse midline glioma, H3 K27M-mutant (that is, with a histone 3 Lys27Met amino acid substitution), is also discussed below. Two aspects of the new WHO integrated diagnosis are particularly important: oligoastrocytomas are no longer recognized as a separate entity and instead, via *IDH* mutation and 1p19q co-deletion status, reflect the genetic profile of either astrocytoma (1p19q intact) or oligodendroglioma (1p19q co-deleted)[36]; and, when histology and molecular features are discordant, molecular features often become the primary determinant of classification.

The five new principal groups have accurately delineated ages at diagnosis and prognosis[8,11,37–39] (FIG. 1; TABLE 1). Patients with glioblastoma, *IDH*-wild type have, on average, the highest age at diagnosis (median 59 years) and the worst prognosis (median overall survival 1.2 years). Patients with glioblastoma, *IDH*-mutant tend to be younger (median age at diagnosis 38 years) and have a better prognosis (median overall survival 3.6 years) than those with glioblastoma, *IDH*-wild type. Patients with astrocytoma, *IDH*-wild type, regardless of grade II or grade III histology, have a median age at diagnosis of 52 years; their median overall survival (only 1.9 years) is closer to that of patients with glioblastoma, *IDH*-wild type than to that of patients with glioblastoma, *IDH*-mutant. By contrast, patients with astrocytoma, *IDH*-mutant have a median age at diagnosis of 36 years and median overall survival of 9.3 years[37,38]. Patients with oligodendroglioma, *IDH*-mutant and 1p19q co-deleted are also relatively young at diagnosis (median age 44 years) and have the longest median overall survival (17.5 years).

Population-based incidence and survival data for the five principal diffuse glioma classes included in the WHO 2016 integrated diagnosis are not yet available, but on the basis of calculations relating to their molecular subtypes (that is, *IDH* mutation and 1p19q co-deletion status)[27,40] and the most recent CBTRUS data[21], we estimate that in 2019, ~71% of newly diagnosed diffuse gliomas will be classified as glioblastoma, *IDH*-wild type; 7% as glioblastoma, *IDH*-mutant; 5% as astrocytoma, *IDH*-wild type; 12% as astrocytoma, *IDH*-mutant; and 5% as oligodendroglioma, *IDH*-mutant and 1p19q co-deleted (TABLE 1).

Diffuse midline glioma, H3 K27M-mutant

The seventh group of the WHO 2016 integrated diagnosis (referenced above) is the diffuse midline gliomas, H3 K27M-mutant. According to the 2007 WHO classification system, these tumours might meet the histological criteria for any grade of astrocytoma, although their poor clinical outcomes most closely resemble those of patients with glioblastoma. First identified in paediatric diffuse intrinsic pontine gliomas, this subgroup is defined by gain-of-function mutations in genes encoding histone H3 (*H3F3A, HIST1H3B, HIST1H3C* or *HIST1H3I*) that result in a Lys to Met amino acid substitution at position 27 (REFS[41,42]). This alteration, which appears early and homogeneously, was also identified in adults with diffuse glioma, predominantly those with young ages of onset and tumours in midline locations such as the spinal cord, thalamus, brainstem and cerebellum. Although the histological classification of these tumours is typically low

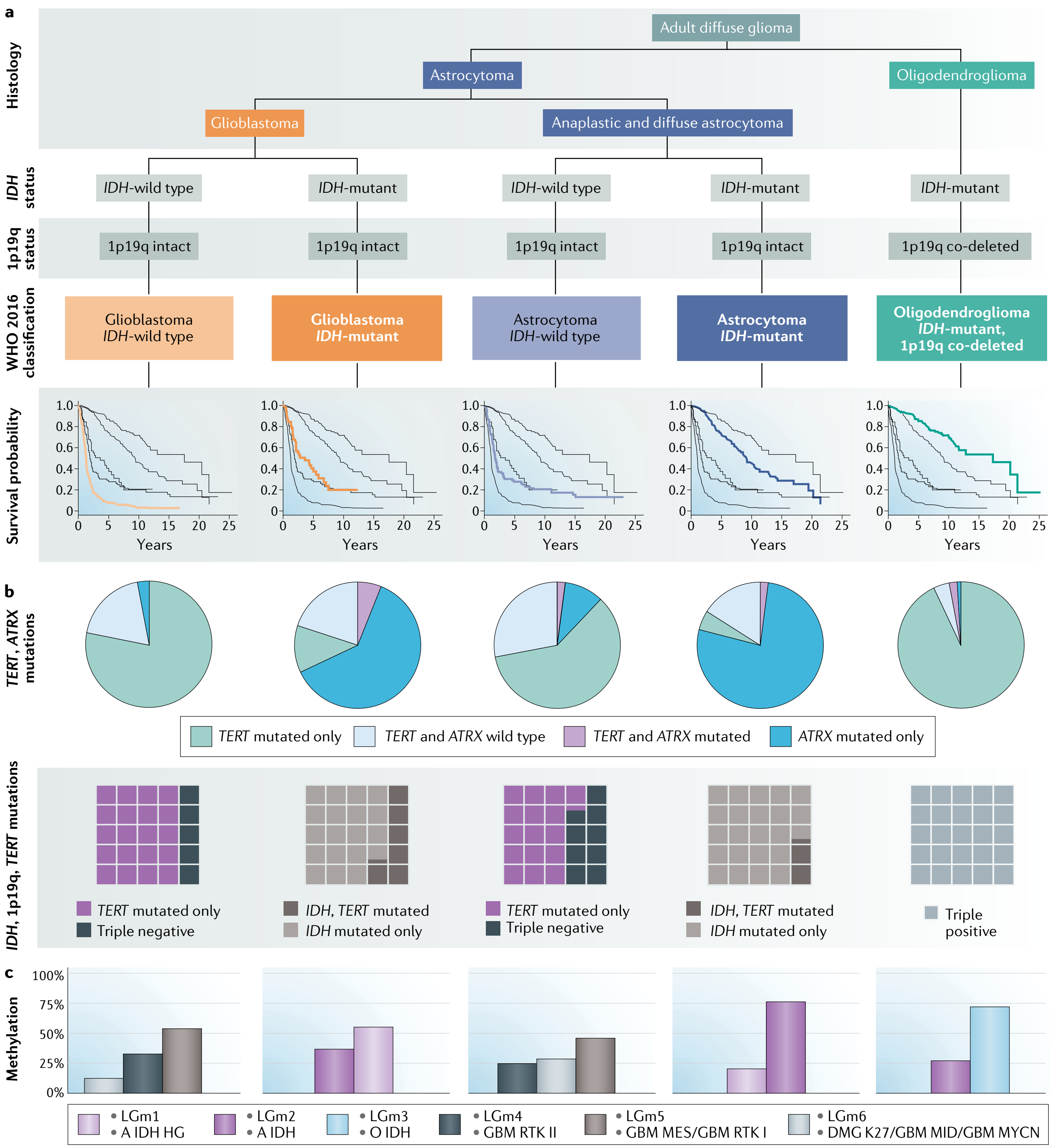

Fig. 1 | **The principal molecular subtypes included in the WHO 2016 classification of newly diagnosed adult diffuse glioma. a** | Histological assessment integrated with molecular diagnosis (based on *IDH* mutation and 1p19q co-deletion status) defines the five principal subtypes of adult diffuse glioma included in the WHO 2016 classification[12,26]. Kaplan–Meier survival curves for each subtype are based on data from 1989 to 2012 (REF.[37]). **b** | For each disease subtype, mutation frequencies are depicted in pie charts showing the proportion of tumours with mutations in *TERT* and/or *ATRX*[37] and in waffle plots showing the relative proportions of *IDH*-mutant, *TERT*-mutant and 1p19q co-deleted tumours in each WHO 2016 disease subtype[27]. **c** | Graphs showing methylation classifications for each WHO 2016 subtype. The methylation categories depicted[49] overlap to a great extent with those identified in a subsequent study[46]. A IDH, *IDH*-mutant astrocytoma; A IDH HG, *IDH*-mutant high-grade astrocytoma; DMG K27, H3 K27-mutant diffuse midline glioma; GBM MES, *IDH*-wild type glioblastoma, mesenchymal subclass; GBM MID, *IDH*-wild type glioblastoma, midline subclass; GBM MYCN, *IDH*-wild type glioblastoma, *MYCN*-amplified subclass; GBM RTK I, *IDH*-wild type glioblastoma, receptor tyrosine kinase I subclass; GBM RTK II, *IDH*-wild type glioblastoma, receptor tyrosine kinase II subclass; O IDH, *IDH*-mutant oligodendroglioma. WHO 2016 classification from REFS[12,26]. Survival and mutation data from REF.[37] with additional mutation data from REF.[27]. Methylation profile data from REFS[46,49].

Table 1 | **Characteristics of the five main WHO 2016 subtypes of adult diffuse glioma**

Characteristic	Glioblastoma, *IDH*-wild type	Glioblastoma, *IDH*-mutant	Astrocytoma, *IDH*-wild type	Astrocytoma, *IDH*-mutant	Oligodendroglioma, *IDH*-mutant and 1p19q co-deleted
Frequent molecular aberrations[a]	• *EGFR* amplification • *CDKN2A* or *CDKN2B* deletion • *PTEN* mutation • *PDGFRA1* mutation • *TP53* mutation • *NF1* mutation • Chromosome 7 gain • Chromosome 10 loss • *MDM2* amplification	• *TP53* mutation • Chromosome 10q loss • *CDKN2A* or *CDKN2B* deletion	• *PTEN* loss • *EGFR* amplification • *NF1* mutation • *TP53* mutation • *PIK3CA* mutation	• *TP53* mutation • *CDKN2A* or *CDKN2B* deletion • *MYCN* amplification • *RTK* or *RIS* mutation	• *CIC* mutation • *FUBP1* mutation • *NOTCH1* mutation • *PI3K* mutation • *CDKN2A* or *CDKN2B* deletion
MGMT promoter methylation (%)[b]	~40	~90	55	~85	~65–100
Median age at diagnosis (years)[c]	59	38	52	36	44
Median overall survival (years)[c]	1.2	3.6	1.9	9.3	17.5
Proportion of 2019 diagnoses (%)[d]	71	7	5	12	5

Data in this table complement the information presented in FIG. 1. [a]Data from REFS[6,8,20,45,46]. [b]Data from REFS[45,60–62]. [c]Data from REF.[37] and in agreement with REFS[8,11,38,39]. [d]Data from the Central Brain Tumor Registry of the United States (CBTRUS) (2011–2015)[21] and REFS[27,40].

grade, their clinical outcome is poor[42]. Consequently, the WHO 2016 classification identified diffusely infiltrating gliomas that arise in the midline and harbour the H3 K27M mutation as a separate entity regardless of the presence of anaplastic histological features[12,43].

Diffuse astrocytoma,* IDH*-wild type

Particular efforts have been made to further characterize tumours in the diffuse astrocytoma, *IDH*-wild type group according to the WHO 2016 classification because the outcomes of patients in this group vary widely[20]. This group includes patients with very low-grade tumours, such as glioneuronal tumours and pilocytic astrocytomas (not discussed in detail in this Review) that have a very favourable course, as well as patients who have a very poor prognosis (similar to that of patients with glioblastoma, *IDH*-wild type). The inclusion of the latter group probably results from sampling of the tumour during resection being insufficient to show histological features of glioblastoma.

Other markers

The Consortium to Inform Molecular and Practical Approaches to CNS Tumor Taxonomy (cIMPACT-NOW) is a working group formed to incorporate advances in defining further subtypes of the WHO 2016 classification system into clinical practice[20,43,44]. For example, cIMPACT-NOW has recommended that diffuse astrocytoma, *IDH*-wild type tumours should undergo further characterization: testing for *EGFR* amplification, gain of chromosome 7, loss of chromosome 10 and the presence of *TERT* promoter mutation. Regardless of tumour histology, if any of these molecular alterations are found, the patient is now considered to have diffuse astrocytic glioma, *IDH*-wild type with molecular features of glioblastoma WHO grade IV[20].

Similarly, several other tumour markers that are not incorporated into the formal WHO 2016 classification system are being used to fine-tune the prognosis of some categories of glioma. Below, we summarize the most promising additional markers. Additional genetic alterations and pathway abnormalities are reviewed elsewhere[6,45] and summarized in TABLE 1.

Telomerase alterations. Numerous studies have shown the classification and prognostic importance of mutations in telomere maintenance genes, including *TERT* and *ATRX*[7,27,46–48]. Typically, *TERT* and *ATRX* alterations are mutually exclusive, probably because of functional redundancy[7,27,46]. A study of over 1,200 adult patients with diffuse glioma aimed to characterize the distribution and additional prognostic value of *TERT* and *ATRX* alterations within the five WHO 2016 groups[37] (FIG. 1). In the glioblastoma, *IDH*-wild type subgroup, >75% of tumours had a *TERT* mutation. Approximately 3% of glioblastoma, *IDH*-wild type tumours had alterations in *ATRX*, which were associated with improved survival (adjusted HR 0.36; 95% CI 0.17–0.81). In the astrocytoma, *IDH*-wild type subgroup, >60% of tumours had a *TERT* mutation whereas 12% had alterations in *ATRX*; *TERT*-wild type tumours were associated with improved survival (adjusted HR 0.48; 95% CI 0.27–0.87). Both the astrocytoma, *IDH*-mutant and glioblastoma, *IDH*-mutant subgroups showed high proportions of *ATRX* mutations (63% and 78%, respectively) and low proportions of *TERT* mutations (5% and 12%, respectively), although neither *TERT* nor *ATRX* alterations were associated with survival in these two groups. Among patients with oligodendroglioma, *IDH*-mutant and 1p19q co-deleted, *TERT* mutations were seen in 94% of tumours and *ATRX* alterations in 3%; *TERT*-wild type tumours were associated with markedly poor survival (adjusted HR 2.46; 95% CI 0.94–6.42). The authors concluded that in some subgroups, additional testing for *ATRX* and/or *TERT* mutations might be warranted.

Methylation signatures. A tumour's methylome includes both somatically acquired DNA methylation changes and characteristics derived from the cell of origin. Distinct DNA methylation signatures are associated with prognosis in glioma[9,11,46,49,50]. For example, mutations in *IDH* cause aberrant methylation of DNA and histones. Hypermethylation of CpG islands is referred to as the glioma CpG-island methylator phenotype (G-CIMP), which is associated with a favourable prognosis[11,25,51]. A study that used both methylation and gene expression analyses to examine over 1,100 newly diagnosed diffuse gliomas included in TCGA found six distinct pan-glioma DNA methylation and transcriptome subtypes, which they termed LGm1–LGm6 (REF.[46]) (FIG. 1).

The *IDH*-mutant gliomas were separated into three groups: LGm1 with 1p19q co-deletion (18%); LGm2 (G-CIMP-high) with 1p19q-intact and highly methylated (25%); and LGm3 (G-CIMP-low) with 1p19q intact and low methylation levels (3%). Interestingly, the LGm1 and LGm2 groups had a similar prognosis; only the LGm3 group had distinctly improved prognosis. The *IDH*-wild type gliomas were separated into three subgroups: LGm4 glioblastomas with classic gene expression profiles (18%); LGm5 glioblastomas with mesenchymal gene expression profiles (27%); and LGm6 glioblastomas with a distinct methylation pattern and stable telomeres (9%). A subset of low-grade gliomas in LGm6 that were similar to pilocytic astrocytoma (that is, with a young age at onset and favourable survival) were referred to as 'pilocytic astrocytoma-like'[46]. Of note, only the pilocytic astrocytoma-like subset showed a survival advantage. All other *IDH*-wild type tumours had indistinguishable survival.

DNA methylation profiling has repeatedly been shown to be a robust method of classifying tumours[23,52]. Accordingly, DNA methylation-based classification has been used to circumvent the interobserver variability highlighted in the study mentioned above[46] and thereby to improve diagnostic precision[49]. The reference data set comprised 2,800 CNS tumours including ~100 tumour types, which were divided into 82 CNS tumour methylation classes. Eight of these classes were subgroups of glioblastoma, which suggested a possible future role of DNA methylation profiling in differentiation of the two glioblastoma subgroups included in the WHO 2016 classification. The 6 pan-glioma methylation classes used in TCGA and shown in FIG. 1 (LGm1–LGm6) overlapped with 9 of the 82 CNS tumour methylation classes, which were named A IDH HG (*IDH*-mutant high-grade astrocytoma), A IDH (*IDH*-mutant astrocytoma), O IDH (*IDH*-mutant oligodendroglioma), GBM RTK II (*IDH*-wild type glioblastoma, receptor tyrosine kinase II subclass), GBM MES (*IDH*-wild type glioblastoma, mesenchymal subclass), GBM RTK I (*IDH*-wild type glioblastoma, receptor tyrosine kinase I subclass), DMG K27 (H3 K27M-mutant diffuse midline glioma), GBM MID (*IDH*-wild type glioblastoma, midline subclass) and GBM MYCN (*IDH*-wild type glioblastoma, *MYCN*-amplified subclass). LGm1, LGm2 and LGm3 show high overlap with A IDH HG, A IDH and O IDH, respectively. LGm4 shows the highest overlap with GBM RTK II. LGm5 shows the highest overlap with GBM MES and GBM RTK I. LGm6 shows the highest overlap with DMG K27, GBM MID and GBM MYCN.

In the validation data set, which included over 1,000 prospectively collected CNS tumours, the DNA methylation-based classifier was concordant with histopathology in 76% of tumours and another 12% (129 tumours) were reclassified on the basis of their methylation profile. In 70% of the 129 reclassified tumours, the WHO 2016 classification was either upgraded or downgraded and several astrocytoma, *IDH*-wild type tumours were reclassified as glioblastoma, *IDH*-wild type. Thus, this DNA methylation classifier could substantially reduce tumour misclassification and decrease the subset of tumours for which diagnosis is difficult because the histology and molecular features are discordant.

MGMT. The DNA damage induced by temozolomide can be repaired by the enzyme methylated-DNA–protein-cysteine methyltransferase (MGMT; also known as 6-*O*-methylguanine-DNA methyltransferase). Accordingly, high levels of *MGMT* mRNA and MGMT protein have been linked to DNA-alkylating agent resistance[53], whereas methylation of CpG islands in the promoter region of *MGMT* (which suppresses its transcription) increases chemosensitivity to these agents[54]. The importance of *MGMT* promoter methylation as a prognostic biomarker and as a predictive biomarker has been shown in multiple clinical trials and research studies[55–59]. A conclusion drawn from several of those studies is that patients with glioblastoma over 65–70 years of age whose tumours lack *MGMT* promoter methylation clearly derive a reduced benefit from temozolomide.

Several studies have examined the interaction between *MGMT* promoter methylation and *IDH1* mutation status because *IDH1* mutation is associated with the globally hypermethylated G-CIMP phenotype[11,51]. Among patients with glioblastoma, *MGMT* promoter methylation rates range from ~40% in *IDH*-wild type tumours to ~90% in *IDH*-mutant tumours (a similar pattern is seen in the limited available data on non-glioblastoma astrocytomas[60,61]; TABLE 1). In grade III and IV gliomas, *MGMT* promoter methylation is a prognostic indicator of improved survival and decreased progression in patients with *IDH*-mutant glioma[62] and predicts a favourable therapeutic response to alkylating agents in patients with *IDH*-wild type glioma[63,64]. In a study that further explored the survival benefit associated with *MGMT* promoter methylation in patients with glioblastoma, *IDH1*-wild type, this benefit was seen only in the 75% of patients whose tumour also harboured a *TERT* promoter mutation[65] (FIG. 1), which suggested that both markers should be included in future risk stratification. Thus, the research to date supports testing for *MGMT* promoter methylation in clinical practice, in particular for two groups: elderly patients with glioblastoma and those with a diagnosis of anaplastic (*IDH*-wild type) glioma[61]. Several studies in older patients (variably defined as age >60 or >65 years) have confirmed *MGMT* promoter methylation as a favourable prognostic factor and as predictive of the response to temozolomide[58,66,67]. Given the increased toxicity of combined radiotherapy and chemotherapy in elderly patients, individuals with *MGMT* promoter methylation might be treated with chemotherapy alone[67] or patients without *MGMT* promoter methylation might be treated with radiotherapy alone[58,66,67].

Historically, *MGMT* promoter methylation testing was not widespread owing to difficulties with the assays and their standardization as well as the existence of few therapeutic alternatives to the current standard of care[54]. However, a data set including over 4,000 patients with glioblastoma pooled from the control arms of four large clinical trials, all of whom received radiation and temozolomide and underwent centralized *MGMT* promoter methylation testing using quantitative methylation-specific PCR, confirmed that patients without *MGMT* promoter methylation had a worse prognosis than patients with *MGMT* promoter methylation. This study also identified that 10% of patients had low *MGMT* promoter methylation. The outcomes of these 'grey zone' patients were aligned more closely with the methylated than with the nonmethylated group. These data support those from ongoing studies showing that both patients with *MGMT* promoter methylation and those with indeterminate *MGMT* promoter methylation status should receive treatment with temozolomide[68]. Perhaps this agent should be withheld only in patients who truly lack *MGMT* promoter methylation.

CDKN2A *and/or* CDKN2B. Aberrations in signalling pathways affecting the cell cycle are a common mechanism of uncontrolled cell growth in glioma. In glioblastoma, this dysregulation most frequently manifests as homozygous deletions in *CDKN2A* and/or *CDKN2B* (50–60%), copy number alterations in other cyclin-dependent pathway genes (~20%), or mutations in retinoblastoma-associated protein (*RB1*) pathway genes (7–10%)[6,10].

Although 50–60% of patients with glioblastoma have loss of *CDKN2A* and/or *CDKN2B*, these mutations were not initially associated with prognosis[6–8]. However, homozygous deletions in *CDKN2A* and/or *CDKN2B* have now been identified as a histology-independent adverse prognostic marker in *IDH*-mutant astrocytomas[69], including in glioblastoma, *IDH*-mutant[70,71]. The loss of *CDKN2A* and/or *CDKN2B*, regardless of WHO 2016 classification, was seen in 11–18% of all tested astrocytomas, *IDH*-mutant; these patients' clinical outcomes were similar to those of patients with glioblastoma, *IDH*-mutant. Although homozygous deletion of *CDKN2A* and/or *CDKN2B* was the strongest predictor of decreased survival, the effect was diminished when combined with other aberrations in the cell cycle pathway, such as amplification of *CDK4* or *CDK6* or deficiency of *RB1*. The adverse effect of *CDKN2A* and/or *CDKN2B* deletion on prognosis might be unique to *IDH*-mutant astrocytomas with *TP53* mutations and might not be seen in oligodendrogliomas or *IDH*-wild type astrocytomas, including glioblastoma[38,69].

Incidence, risk and survival

Globally, the incidence of adult diffuse gliomas differs substantially between countries[72–78]. In adults over 40 years of age, the age-adjusted annual incidence of astrocytic tumours is 6.8 cases per 100,000 people[79]. Countries with predominantly northern European populations have a higher rate (ranging from 7.8 in the United States to 9.6 in Australia and New Zealand) than countries with predominantly Asian or African populations (ranging from 1.9 in Southeast Asia to 3.3 in India)[79]. Similar patterns in incidence are seen in oligodendroglial tumours. The wide range of incidence values is confounded by differences in access to medical imaging, case ascertainment and surveillance. Nonetheless, the large difference in incidence between European and Asian populations is observed both when comparing different countries and within the United States and United Kingdom[79,80]. The fourfold higher incidence in western Europe (8.5 cases per 100,000) than in prosperous East Asian countries (such as Japan and Singapore: 1.9 cases per 100,000) suggests that ethnic, cultural and/or environmental differences might be more likely to explain the observed differences in incidence than reporting bias[3,78,79]. Future studies comparing genetic, environmental and regional lifestyle risk factors within and between countries will be important to understand these differences[78].

The latest incidence data in the United States for adult diffuse glioma are derived from the 2018 CBTRUS report[21], which relies on the WHO 2007 classification system as the WHO 2016 system has not yet been used widely enough in population data sources. As the CBTRUS definition of glioma includes pilocytic astrocytoma, unique astrocytoma variants, ependymal tumours and malignant glioma not otherwise specified, when citing CBTRUS data, we have included only data on diffuse glioma in adults. However, other references cited contain various definitions of glioma, which sometimes include paediatric patients. Therefore, in this article, we differentiate between adult diffuse glioma and glioma. As noted above, the WHO 2016 classification does not include the prior histological category of oligoastrocytoma. However, CBTRUS and other reporting registries retain that category and have not yet published population data based on the new classification. Therefore, in this section we use the older WHO 2007 classification used by registries for population statistics and, when available, limited data from studies that used the WHO 2016 subtypes.

In the United States, diffuse gliomas account for 72% of malignant brain tumours in adults and 21% of all primary brain and other CNS tumours in adults[21,81]. Glioblastoma, the most rapidly fatal form, accounts for 73% of diffuse gliomas, 52% of malignant non-metastatic brain tumours and 15% of all primary brain and other CNS tumours in adults[21]. Although diffuse gliomas account for <1% of all new cancers[21,82], they are associated with considerable morbidity and mortality. Since 2000, the incidence of glioma has been relatively stable, with a small but statistically significant increase from 2000 to 2008 followed by a statistically significant decrease from 2008 to 2014 (REFS[21,72,83,84]). In 2019, new cases of adult diffuse glioma in the United States are estimated to approach 17,000 whereas during 1990–1994, <10,000 cases annually were reported[85]. The large increase observed after 1994 is likely to be due to improved detection through advances in diagnostic imaging.

Fig. 2 | **Annual average age-adjusted incidence and relative survival data for non-glioblastoma and glioblastoma CNS tumours.** The Central Brain Tumor Registry of the United States (CBTRUS) and other global reporting registries have not yet published glioma statistics based on the WHO 2016 classification system. Accordingly, this figure presents the latest available published CBTRUS statistics from 2011 to 2015, as well as data accessed directly from CBTRUS, which are based on the WHO 2007 classification system. Note that different scales are used for the axes describing the incidence of glioblastoma versus the other glioma types. Data from REF.[21].

Most population-based studies show that, in addition to the geographical differences, glioma incidence varies by age, sex, ethnicity and tumour histology (or WHO 2016 subtype) and that survival for patients with glioma also consistently varies by age, sex and tumour subtype.

Age and sex

Within diffuse glioma subtypes, population incidence varies according to age and sex (FIG. 2). Astrocytoma (including glioblastoma and diffuse or anaplastic astrocytoma) incidence increases with age and peaks between the ages of 75 years and 84 years. Men have a 40–50% higher incidence than women at all ages[21,86] (FIG. 2). In 2000–2014, 1-year and 5-year relative survival for adults with glioblastoma was 41% and 5%, respectively. Corresponding 1-year and 5-year relative survival for those with non-glioblastoma astrocytomas was 72% and 44%, respectively[81] (FIG. 2). In a large study that used the WHO 2016 classification, the median age at diagnosis was lower for patients with *IDH*-mutant astrocytomas (36 years) and glioblastomas (38 years) than for those with *IDH*-wild type astrocytomas (52 years) and glioblastomas (59 years)[37] (TABLE 1). Median overall

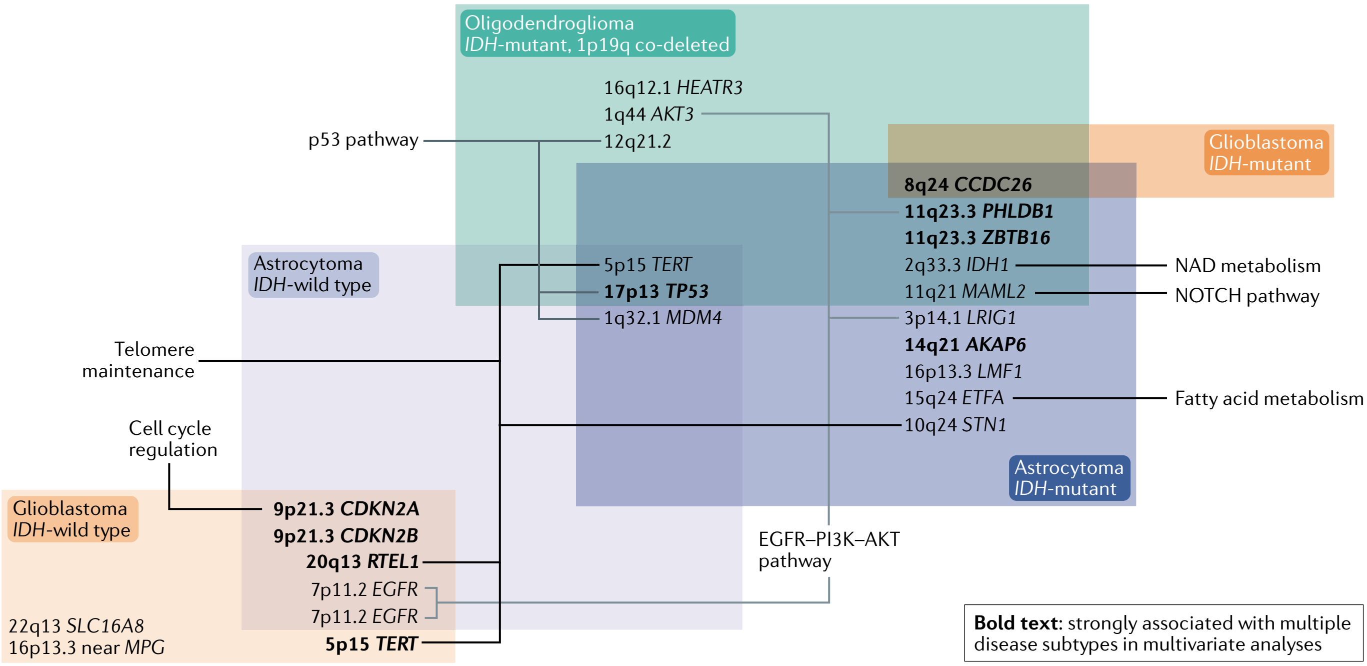

Fig. 3 | **Heritable germline risk factors for the WHO 2016 subtypes of adult glioma.** Summary of the relationship between heritable germline risk factors and WHO 2016 classification. Overlapping boxes contain alterations shared by different diffuse glioma tumour subtypes. Involved genes are known to be relevant to the indicated biological pathways but, with a few exceptions, the functional consequences of individual single-nucleotide polymorphisms are still unknown. Data from REFS[19,126].

survival is better in patients with *IDH*-mutant astrocytomas (9.3 years) and *IDH*-mutant glioblastomas (3.6 years) than in those with *IDH*-wild type glioblastoma (1.9 years) and *IDH*-wild type astrocytoma (1.2 years)[37] (TABLE 1).

The population incidence of grade II oligodendrogliomas and oligoastrocytomas peaks at age 35–44 years (FIG. 2). Men have an ~25–60% higher incidence than women depending on age. In 2000–2014, 1-year and 5-year relative survival for oligodendroglioma was 90% and 70%, respectively. The incidence of anaplastic oligoastrocytic tumours peaks between the ages of 55 years and 64 years; in this age group, men show a 20% higher incidence than women (range 20–40% across all age groups). The median age at diagnosis is 44 years for those with *IDH*-mutant and 1p19q co-deleted oligodendrogliomas and median overall survival is 17.5 years[37] (TABLE 1).

Similar age and sex patterns for astrocytomas (including glioblastoma) are also seen in Europe[87,88]. These age patterns may be related to the time needed to acquire the multiple genetic alterations required for malignant transformation. As the world population continues to grow and age, the incidence of diffuse glioma is expected to increase. As yet, no comprehensive explanations account for the peaks in glioma incidence observed in particular age groups nor for the increased incidence of glioma in men. The lack of definitive associations between hormone exposure and glioma risk in multiple large studies[86,89–92], as well as the fact that most cancers show increased incidence in men, suggest the existence of unidentified risk factors.

Ethnicity

In the United States, the incidence of adult diffuse glioma is highest in non-Hispanic white individuals (versus Hispanic white, black, American Indian or Alaska Native and Asian or Pacific Islander people). Non-Hispanic white people have over a twofold higher rate of glioblastoma than do black, Asian or Pacific Islander and American Indian or Alaska Native individuals and a 30% higher rate than Hispanic white individuals[81]. This pattern is also consistently seen in other diffuse glioma subtypes, for which incidence rates are similarly highest in non-Hispanic white individuals. Data from the United States also suggest that the lifetime risk of developing a malignant brain tumour is 25% higher in non-Hispanic white than in Hispanic white individuals and is twice as high as the risk in black individuals[81]. In glioblastoma, 5-year relative survival is lowest (5%) in non-Hispanic white individuals and highest (9%) in Asian or Pacific Islander individuals. For non-glioblastoma astrocytoma, 5-year relative survival is lowest in non-Hispanic white people (43.2%) and higher in all other ethnic groups (44.1%–50.8%), whereas for oligodendroglioma 5-year relative survival is lowest in black individuals (63.8%)[81].

Additional survival factors

Additional factors associated with survival are the extent of tumour resection, Karnofsky performance score and treatment. Multiple studies have indicated that complete resection is beneficial[93–95] even in low-grade tumours, for which early resection is preferable to watchful waiting or biopsy alone[96]. Although difficult to prove, the influence of extent of tumour resection on survival could be confounded by tumour location, whether it is resectable or non-resectable and/or by clinical judgement[97]. After 2005, when clinical trial results showing an increase in median overall survival from 12.1 months to 14.6 months were published, the standard of care for patients with newly diagnosed glioblastoma has been treatment with the DNA-alkylating agent temozolomide, radiotherapy and surgery (termed the Stupp protocol)[98]. Among patients with glioblastoma treated according to this protocol, median survival in contemporary clinical trials is 14–17 months[57,72,99–104]. In one trial, for which preliminary data were published in 2015 (REF.[97]) and the complete report in 2017 (REF.[98]), the addition of tumour-treating fields (an antimitotic treatment modality) to maintenance temozolomide resulted in median overall survival of 21 months (versus 16 months in the temozolomide-alone group) and 5-year overall survival of 13%. Although survival in glioblastoma remains poor, the conditional probability of surviving for ≥1 year past 2 years has repeatedly been shown to be much more favourable than surviving for ≥1 year past the initial diagnosis[100,105,106].

Inherited genetic risk factors

Single-gene hereditary cancer syndromes such as Li–Fraumeni syndrome, neurofibromatosis, Lynch syndrome, melanoma–neural system tumour syndrome, Ollier disease and tuberous sclerosis cause a small percentage of diffuse gliomas in adults[40,107]. Gliomagenesis in these familial cancer syndromes might differ from that in gliomas presumed to be spontaneous[108]. Gliomas arising in patients with neurofibromatosis type 1 lack mutations in *IDH1, IDH2* and the H3 K27M mutation. In addition, high-grade tumours in patients with neurofibromatosis type 1 often harbour loss of *ATRX* whereas low-grade tumours in these patients often show copy number gain of *TERT*. The methylation signature of these tumours falls into the LGm6 cluster and this subgroup can probably be subdivided further on the basis of *ATRX* status[109]. Gliomas in patients with germline *TP53* mutations and Li–Fraumeni syndrome show enrichment of the *IDH1* R132C mutation (which accounts for < 5% of all *IDH* mutations) and suggest a pathological relationship with *TP53* mutations[110–112].

In the large majority of patients with glioma who do not have these familial cancer syndromes, 5–10% have a family history of glioma[113–115]. In fact, first-degree relatives of patients with glioma have a twofold increased risk of developing a primary brain tumour compared with first-degree relatives of people who do not have glioma[114–116]. Possible reasons for the observed familial clustering of glioma are shared genetic and/or environmental factors. Given the paucity of identified environmental risk factors (described below) and the likelihood that glioma familial aggregation is a polygenic effect[117,118], linkage analysis, whole-exome sequencing and genome-wide association studies (GWAS) are ongoing to improve our understanding of how germline variants contribute to the risk of glioma. A potential risk locus for familial glioma was detected on chromosome 15 in an early, small linkage analysis study of a high-risk family[16]. A subsequent larger study identified a single locus on chromosome 17 and three other loci on chromosomes 6, 12 and 18 (REFS[17,18]). Whole-exome sequencing of *POT1*, which modulates telomerase activity, led to the identification of germline DNA mutations linked to oligodendroglioma in multiple affected families[119].

A large GWAS including >12,000 patients with glioma and 18,000 controls[19] identified or validated a total of 25 single-nucleotide polymorphisms (SNPs) that were strongly associated with glioma risk in adults[17,19,120–122]. Of these 25 SNPs, 11 were associated with glioblastoma risk and 19 were associated with the risk of non-glioblastoma glioma (5 SNPs were associated with risk of both glioblastoma and non-glioblastoma gliomas). The strongest relative risk is for variants at 8q24.21, which confer more than a sixfold relative risk of developing astrocytoma, *IDH*-mutant or oligodendroglioma[40,123,124].

These 25 risk loci were subsequently included in a model to assess the relative risk and lifetime risk of developing glioma[125]. In another study, these 25 risk loci were studied in over 1,600 tumours for associations with the *IDH* mutation, *TERT* promoter mutation and 1p19q co-deletion molecular subgroups[126]. The association of 8q24.21 variants with oligodendrogliomas was validated[126]. These same researchers also examined univariate and multivariate associations between these risk loci and the WHO 2016 glioma subtypes. The majority of the SNPs lie in or near genes known to be involved in specific pathways, summarized in FIG. 3. For example, the SNP at 17p13.1 lies in the 3′ untranslated region of *TP53* (this SNP alters the polyadenylation signal of *TP53*, thereby impairing processing of *TP53* mRNA) and is associated with the risk of several types of cancer.

Telomere maintenance-related genes are important in glioma prognosis and classification. SNPs near *TERT* and *RTEL1* (encoding regulator of telomere elongation helicase 1), both of which encode proteins involved in telomere maintenance, were associated with gliomas that had *TERT* promoter mutations, reinforcing the

Box 1 | **Non-genetic risk factors for glioma**

Association with increased risk[a]

- Ionizing radiation from atomic bombs
- Therapeutic radiation exposure to the brain during childhood
  - Radiation in childhood for treatment of tinea capitis
  - Radiation for childhood leukaemia
  - CT scans in childhood to doses >50 mGy

Consistently reported association with decreased risk

- Allergy and atopic conditions — asthma, hay fever and eczema

Possible but unconfirmed association with increased risk

- Non-ionizing radiation, such as radiofrequency electromagnetic fields

No clear association

- Pesticide or chlorinated solvent exposure[158–160]
- Obesity and/or high BMI
- Cigarette use[161–165]
- Traumatic brain injury[166–169]
- Alcohol consumption[164]

[a]References for this section are presented in the accompanying article. Non-genetic risk factors for adult glioma have been thoroughly reviewed elsewhere[72].

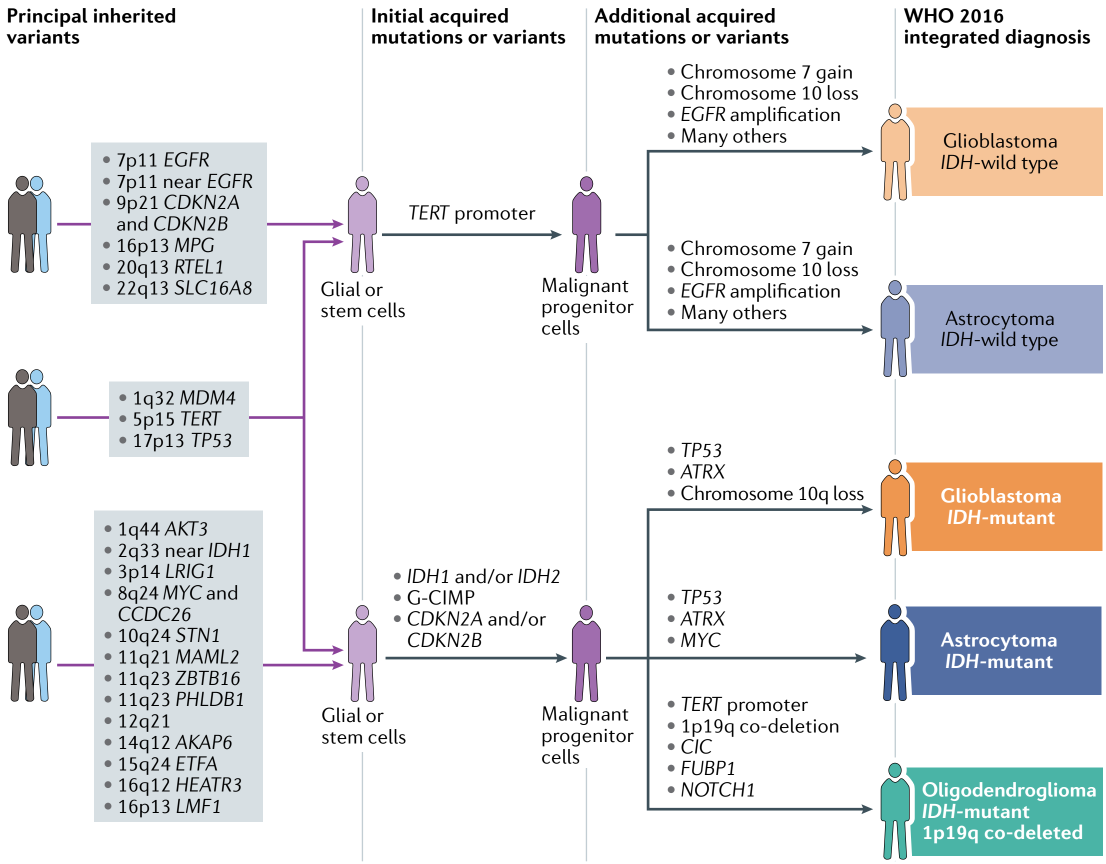

Fig. 4 | **Hypothesized pathways of glioma development.** Our current understanding of gliomagenesis involves an interplay of inherited risk variants (FIG. 3) and acquired alterations (FIG. 1, TABLE 1). Several additional single-nucleotide polymorphisms are associated with glioma but not included in this figure: 1p31 in *RAVER2*; 10q25 in *VTI1A*; and 11q14. G-CIMP, glioma CpG-island methylator phenotype. Data from REFS[17,19,120,122,126].

relationship between SNP genotype and biology[126]. SNPs near *TERT*, *STN1* and *RTEL1* are associated with 1–4 of the five principal WHO 2016 groups of diffuse glioma (FIG. 3). Only glioblastoma, *IDH*-mutant is not associated with these SNPs, probably owing to the small sample size of this study ($n = 68$) and, therefore, the lack of statistical power[126]. One of these SNPs, in *TERT*, results in alternative splicing, which results in lack of telomerase activity[127].

Given the results of these GWAS as well as a Mendelian randomization study showing that genetically increased telomere length is associated with a fivefold increase in the risk of glioma[128], interference with telomere maintenance is clearly a hallmark of gliomagenesis. However, with the exceptions of the *TP53* SNP and the telomere-associated SNPs mentioned above, the reader should note that neither the actual functional effects of the glioma-risk-associated SNPs nor the developmental window within which these SNPs act to increase glioma risk is known.

Non-genetic risk factors

Very few non-genetic risk factors for development of adult diffuse gliomas have been established, despite decades of research[129]. Only ionizing radiation exposures are considered causal of glioma, but they account for an overall small number of cases. Thus, the specific aetiology of most diffuse gliomas remains unclear. Eventually, now that more is known about genetic risk factors, understanding the interplay between environmental exposures, developmental factors and genetic susceptibilities and how these contribute to tumour formation might be possible[130]. BOX 1 summarizes the evidence for and against several non-genetic risk factors, some of which are discussed in more detail below.

Radiation

Exposure to ionizing radiation, particularly in childhood, remains the strongest environmental risk factor for diffuse glioma. Ionizing radiation damages DNA, which can lead to oncogenesis as early as 7–9 years after exposure[130]. The risk of diffuse glioma is dose-dependent, and both low doses of ionizing radiation, such as in atomic bomb survivors[131] and the higher doses of therapeutic radiation used to treat childhood infections and cancers increase glioma risk. For example, therapeutic radiation is associated with a 3–7-fold increase in glioma risk[129,132–134].

The contributions of various types of non-ionizing radiation exposures to glioma risk (including extremely low-frequency exposures, microwaves and radiofrequencies used in cell phones) have been widely studied, but the results of these investigations are inconclusive because of difficulties in establishing lifetime exposures to such radiation and the lack of any convincing biological explanation of the carcinogenic effects of these exposures[129,135–138].

Immune, allergic and atopic conditions

Numerous studies have suggested that a history of allergies or other atopic conditions (defined as hay fever, eczema and asthma) are associated with decreased glioma risk[139,140]. The largest study on this topic confirmed a markedly reduced risk of glioma in individuals with a history of allergy[139]. Moreover, a Mendelian randomization study of 12,488 patients with glioma and 18,169 controls reported that SNPs associated with atopic dermatitis were associated with a reduced risk of glioma[141]. However, the same study did not find any correlations between glioma risk and SNPs previously associated with other atopic conditions, such as asthma, hay fever and elevated IgE levels[141]. These conflicting results highlight the complexity of the relationship between glioma risk, atopy and allergy.

One hypothesis proposed to explain this inverse association is that the atopic state translates into heightened immunosurveillance, which enables the early detection and suppression of tumour growth[140,142,143]. Another hypothesis, supported by the inverse relationship between IgE levels and glioma risk, is the increased ability of an overactive immune system to eradicate potential environmental toxins, particularly respiratory allergens, which might lead to tumour formation[144]. However, as some of the medications given to patients with brain tumours (such as steroids and temozolomide) affect IgE levels, the potential effects of these exposures on the relationship between IgE levels and glioma risk warrant further investigation[141,145].

A report from 1977 first noted that patients with glioma have altered peripheral blood immune cell profiles[146]. Deficiencies in $CD4^+$ T cells and other lymphocytes[147–149], as well as accumulations of myeloid cells[150–154], are observed in patients with glioma and are especially prominent in patients with glioblastoma. Although several treatments for glioma (including steroids, radiation and alkylating agents) can adversely affect peripheral blood counts, glioma-related deficiencies in immune cells can precede and be independent of therapy[155]. The prognostic importance of these changes is under study and is of obvious importance in the development of immunotherapies. Moreover, the aetiological relevance of systemic immune profiles with respect to glioma risk is essentially unknown. New approaches for characterizing peripheral immune profiles on the basis of archival DNA[156,157] might be useful to promote large-scale epidemiological studies that address the aetiological role of systemic immune alterations in gliomagenesis.

Conclusions

Discoveries over the past 15 years have provided tremendous insight into the molecular pathways and inherited genetic risks associated with gliomagenesis, which have provided a solid foundation for future research. The integration of molecular data into the WHO 2016 classification system for diffuse glioma is critical for both improving prediction of prognosis and tailoring treatment. However, more work is needed to understand the interactions between genetic and non-genetic risk factors in gliomagenesis and to identify risk variants within specific molecular subtypes of adult diffuse glioma (as shown in FIG. 4). Furthermore, the role of patients' immune status at various stages of the disease process also remains to be elucidated. To address these critical efforts, future studies will need increased international collaboration and large databases of well-annotated biospecimens that include clinical outcome and imaging data as well as detailed exposure histories.

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